Algorithm 2: MDAV

```
Input: X, k
Output: C: set of clusters
C ← ∅
while |X| ≥ 3k do
    x_c ← mean_record(X)
    x_r ← argmax_{x_i} distance(x_i, x_c)
    x_s ← argmax_{x_i} distance(x_i, x_r)
    C_r ← cluster(x_r, k, X) // Algorithm 3
    C_s ← cluster(x_s, k, X)
    C ← C ∪ {C_r, C_s}
    X ← X \ C_r \ C_s
end
if 2k ≤ |X| < 3k then
    x_c ← mean_record(X)
    x_r ← argmax_{x_i} distance(x_i, x_c)
    C_r ← cluster(x_r, k, X)
    C ← C ∪ {C_r}
    X ← X \ C_r
else
    C ← C ∪ {X}
end
return C
```

Figure 3 depicts the representatives (centroids) of clusters computed by MDAV with $k = 200$ on the example data set of Fig. 1. The figure also shows the decision trees that are obtained as explanations for three of the clusters.

Algorithm 3: cluster

```
Input: x, k, X
Output: C: cluster
C ← {x}
while |C| < k do
    x_i ← argmin_{x_i} distance(x_i, x)
    C ← C ∪ {x_i}
    X ← X \ {x_i}
end
return C
```

4 Empirical Work

We generated a data set consisting of 30,000 records, each with 10 numeric continuous attributes and a single binary class label using the `make_classification`